

Exercise 1

A panel data model was estimated with the dataset crime.dta excerpted from the textbook Principles of Econometrics, R. Carter Hill, William E. Griffiths, Guay C. Lim, Ed. 3, Wiley, 2007. The panel data model had form:

$$lcrmrte_{i,t} = \beta_0 + \beta_1 lprbarr_{i,t} + \beta_2 lprbconv_{i,t} + \beta_3 lprbpris_{i,t} + \beta_4 lavgsen_{i,t} + \beta_5 lpolpc_{i,t} + \beta_6 ldensity_{i,t} + \beta_7 ltaxpc_{i,t} + \beta_8 lwcon_{i,t} + \beta_9 lwfir_{i,t} + u_i + \epsilon_{i,t}, \quad (1)$$

and the data is described at the end of this file.

Apply the general-to-specific procedure to the model. Use Wald test.

Step 1

The general model:

Fixed-effects (within) regression	Number of obs	=	630
Group variable: county	Number of groups	=	90
R-sq: within = 0.3748	Obs per group: min	=	7
between = 0.3317	avg	=	7.0
overall = 0.3331	max	=	7
	F(9,531)	=	35.37
corr(u_i, Xb) = 0.1444	Prob > F	=	0.0000

lcrmrte	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
lprbarr	-.3917945	.0334336	-11.72	0.000	-.457473	-.3261161
lprbconv	-.3111651	.0218356	-14.25	0.000	-.3540598	-.2682704
lprbpris	-.2134135	.033476	-6.38	0.000	-.2791752	-.1476518
lavgsen	.0327006	.0259671	1.26	0.208	-.0183103	.0837115
lpolpc	.4253936	.0276187	15.40	0.000	.3711382	.4796489
ldensity	-.0437532	.2643179	-0.17	0.869	-.5629903	.475484
ltaxpc	-.0214103	.03942	-0.54	0.587	-.0988486	.056028
lwcon	-.0767867	.0381908	-2.01	0.045	-.1518102	-.0017632
lwfir	-.0345552	.0289553	-1.19	0.233	-.0914362	.0223258
_cons	-1.137501	.3002654	-3.79	0.000	-1.727355	-.5476472
sigma_u	.4551673					
sigma_e	.14553932					
rho	.90724387	(fraction of variance due to u_i)				

F test that all u_i=0: F(89, 531) = 33.97 Prob > F = 0.0000

We would like to drop variable ldensity

	Estimate	Std. Error	t-value	Pr(> t)
ldensity	-0.043753	0.264318	-0.1655	0.86859

We drop this variable from model (1), because the variable is insignificant.

Step 2

Model (1) with ldensity dropped

$$lcrmte_{i,t} = \beta_0 + \beta_1 lprbarr_{i,t} + \beta_2 lprbconv_{i,t} + \beta_3 lprbpris_{i,t} + \beta_4 lavgsen_{i,t} + \beta_5 lpolpc_{i,t} + \beta_7 ltaxpc_{i,t} + \beta_8 lwcon_{i,t} + \beta_9 lwfir_{i,t} + u_i + \epsilon_{i,t} \quad (2)$$

Fixed-effects (within) regression Number of obs = 630

lcrmte	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
lprbarr	-.3913002	.0332695	-11.76	0.000	-.456656	-.3259444
lprbconv	-.3107911	.0216985	-14.32	0.000	-.3534164	-.2681659
lprbpris	-.2132688	.033434	-6.38	0.000	-.2789477	-.1475899
lavgsen	.0328589	.0259258	1.27	0.206	-.0180706	.0837883
lpolpc	.4250745	.0275262	15.44	0.000	.3710011	.4791478
ltaxpc	-.0244486	.0348541	-0.70	0.483	-.0929171	.0440198
lwcon	-.0781035	.0373189	-2.09	0.037	-.1514141	-.004793
lwfir	-.0353831	.0284941	-1.24	0.215	-.0913577	.0205916
_cons	-1.11619	.2710159	-4.12	0.000	-1.648582	-.5837969

F test that all u_i=0: F(89, 532) = 40.69 Prob > F = 0.0000

We would like to drop ltaxpc – it is insignificant. To do so, we should verify joint hypothesis $H_0 : \beta_{ldensity} = \beta_{ltaxpc} = 0$ for the general model. Why? Lovell bias. We estimate the general model test this hypothesis for joint hypothesis regarding ldensity and ltaxpc.

Fixed-effects (within) regression Number of obs = 630

lcrmte	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
lprbarr	-.3917945	.0334336	-11.72	0.000	-.457473	-.3261161
lprbconv	-.3111651	.0218356	-14.25	0.000	-.3540598	-.2682704
lprbpris	-.2134135	.033476	-6.38	0.000	-.2791752	-.1476518
lavgsen	.0327006	.0259671	1.26	0.208	-.0183103	.0837115
lpolpc	.4253936	.0276187	15.40	0.000	.3711382	.4796489
ldensity	-.0437532	.2643179	-0.17	0.869	-.5629903	.475484
ltaxpc	-.0214103	.03942	-0.54	0.587	-.0988486	.056028
lwcon	-.0767867	.0381908	-2.01	0.045	-.1518102	-.0017632
lwfir	-.0345552	.0289553	-1.19	0.233	-.0914362	.0223258
_cons	-1.137501	.3002654	-3.79	0.000	-1.727355	-.5476472

F test that all u_i=0: F(89, 531) = 33.97 Prob > F = 0.0000

test ldensity ltaxpc

(1) ldensity = 0

(2) ltaxpc = 0

F(2, 531) = 0.26

Prob > F = 0.7717

These two variables are jointly insignificant, therefore we can remove ltaxpc from model (2).

Step 3

Model (2) with ltaxpc dropped = the general model without ldensity and ltaxpc

$$lcrmte_{i,t} = \beta_0 + \beta_1 lprbarr_{i,t} + \beta_2 lprbconv_{i,t} + \beta_3 lprbpris_{i,t} + \beta_4 lavgsen_{i,t} + \beta_5 lpolpc_{i,t} + \beta_8 lwcon_{i,t} + \beta_9 lwfir_{i,t} + u_i + \epsilon_{i,t} \quad (3)$$

Fixed-effects (within) regression Number of obs = 630

lcrmte	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
lprbarr	-.3899751	.0332	-11.75	0.000	-.4551941	-.3247561
lprbconv	-.3101646	.0216698	-14.31	0.000	-.3527333	-.267596
lprbpris	-.2115777	.0333311	-6.35	0.000	-.2770541	-.1461013
lavgsen	.033955	.0258663	1.31	0.190	-.0168575	.0847674
lpolpc	.4227448	.027312	15.48	0.000	.3690923	.4763972
lwcon	-.0894589	.033609	-2.66	0.008	-.1554812	-.0234367
lwfir	-.040738	.0274393	-1.48	0.138	-.0946404	.0131644
_cons	-1.120211	.2708261	-4.14	0.000	-1.652228	-.5881932

F test that all u_i=0: F(89, 533) = 40.83 Prob > F = 0.0000

We would like to drop insignificant lavgsen variable. To do so, the joint hypothesis $H_0 : \beta_{ldensity} = \beta_{ltaxpc} = \beta_{lavgsen} = 0$ cannot be rejected in the general model. Why? Lovell bias. We estimate the general model test this hypothesis for joint hypothesis regarding ldensity, ltaxpc, lavgsen.

Fixed-effects (within) regression Number of obs = 630

lcrmte	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
lprbarr	-.3917945	.0334336	-11.72	0.000	-.457473	-.3261161
lprbconv	-.3111651	.0218356	-14.25	0.000	-.3540598	-.2682704
lprbpris	-.2134135	.033476	-6.38	0.000	-.2791752	-.1476518
lavgsen	.0327006	.0259671	1.26	0.208	-.0183103	.0837115
lpolpc	.4253936	.0276187	15.40	0.000	.3711382	.4796489
ldensity	-.0437532	.2643179	-0.17	0.869	-.5629903	.475484
ltaxpc	-.0214103	.03942	-0.54	0.587	-.0988486	.056028
lwcon	-.0767867	.0381908	-2.01	0.045	-.1518102	-.0017632
lwfir	-.0345552	.0289553	-1.19	0.233	-.0914362	.0223258
_cons	-1.137501	.3002654	-3.79	0.000	-1.727355	-.5476472

F test that all u_i=0: F(89, 531) = 33.97 Prob > F = 0.0000

test ldensity ltaxpc lavgsen

- (1) ldensity = 0
- (2) ltaxpc = 0
- (3) lavgsen = 0

F(3, 531) = 0.75
 Prob > F = 0.5252

These three variables are jointly insignificant, thus lavgsen might be removed from model (3).

Step 4

We estimate the general model without *ldensity*, *ltaxpc*, *lavgsen*

$$\begin{aligned} lcrmte_{i,t} = & \beta_0 + \beta_1 lprbarr_{i,t} + \beta_2 lprbconv_{i,t} + \beta_3 lprbpris_{i,t} \\ & + \beta_5 lpolpc_{i,t} + \beta_8 lwcon_{i,t} + \beta_9 lwfir_{i,t} + u_i + \epsilon_{i,t} \end{aligned} \quad (4)$$

Fixed-effects (within) regression Number of obs = 630

lcrmte	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
lprbarr	-.3919041	.03319	-11.81	0.000	-.457103	-.3267052
lprbconv	-.3111799	.0216706	-14.36	0.000	-.3537501	-.2686098
lprbpris	-.2105102	.0333437	-6.31	0.000	-.2760112	-.1450093
lpolpc	.4217146	.0273192	15.44	0.000	.3680482	.475381
lwcon	-.0898225	.0336306	-2.67	0.008	-.155887	-.0237581
lwfir	-.0412706	.0274549	-1.50	0.133	-.0952034	.0126622
_cons	-1.051047	.2658314	-3.95	0.000	-1.57325	-.528843

F test that all $u_i=0$: F(89, 534) = 40.89 Prob > F = 0.0000

We would like to drop insignificant *lavgsen* variable. To do so, the joint hypothesis $H_0 : \beta_{ldensity} = \beta_{ltaxpc} = \beta_{lavgsen} = \beta_{lwfir} = 0$ cannot be rejected in the general model. Why? Lovell bias. We estimate the general model test this hypothesis for joint hypothesis regarding *ldensity*, *ltaxpc*, *lavgsen*, *lwfir*.

Fixed-effects (within) regression Number of obs = 630

lcrmte	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
lprbarr	-.3917945	.0334336	-11.72	0.000	-.457473	-.3261161
lprbconv	-.3111651	.0218356	-14.25	0.000	-.3540598	-.2682704
lprbpris	-.2134135	.033476	-6.38	0.000	-.2791752	-.1476518
lavgsen	.0327006	.0259671	1.26	0.208	-.0183103	.0837115
lpolpc	.4253936	.0276187	15.40	0.000	.3711382	.4796489
ldensity	-.0437532	.2643179	-0.17	0.869	-.5629903	.475484
ltaxpc	-.0214103	.03942	-0.54	0.587	-.0988486	.056028
lwcon	-.0767867	.0381908	-2.01	0.045	-.1518102	-.0017632
lwfir	-.0345552	.0289553	-1.19	0.233	-.0914362	.0223258
_cons	-1.137501	.3002654	-3.79	0.000	-1.727355	-.5476472

F test that all $u_i=0$: F(89, 531) = 33.97 Prob > F = 0.0000

test *ldensity* *ltaxpc* *lavgsen* *lwfir*

```
( 1)  ldensity = 0
( 2)  ltaxpc = 0
( 3)  lavgsen = 0
( 4)  lwfir = 0
      F( 4, 531) = 1.12
      Prob > F = 0.3446
```

These three variables are jointly insignificant, thus *lwfir* might be removed from model (4).

Step 5

We estimate the general model without `ldensity`, `ltaxpc`, `lavgsen`, `lwfir`

$$lcrmrte_{i,t} = \beta_0 + \beta_1 lprbarr_{i,t} + \beta_2 lprbconv_{i,t} + \beta_3 lprbpris_{i,t} + \beta_5 lpolpc_{i,t} + \beta_8 lwcon_{i,t} + u_i + \epsilon_{i,t} \quad (5)$$

Fixed-effects (within) regression Number of obs = 630

lcrmrte	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
lprbarr	-.3891623	.0331788	-11.73	0.000	-.454339	-.3239856
lprbconv	-.3101617	.0216855	-14.30	0.000	-.3527609	-.2675625
lprbpris	-.2072818	.0333136	-6.22	0.000	-.2727233	-.1418402
lpolpc	.4193427	.0273057	15.36	0.000	.3657031	.4729823
lwcon	-.1054215	.0320272	-3.29	0.001	-.168336	-.0425071
_cons	-1.204457	.2457499	-4.90	0.000	-1.68721	-.7217035

F test that all u_i=0: F(89, 535) = 41.08 Prob > F = 0.0000

All variables in the model (5) are significant at the 5% significance level. The end of the general-to-specific procedure.

R replication code

```
library("Formula")
library("plm")
library("sandwich")
library("zoo")
library("lmtest")
library("aod")

crime.all = read.csv(file="crime.csv", header=TRUE, sep=",")
View(crime.all)

crime = pdata.frame(crime.all, index=c("county", "year"))
View(crime)

# -----
# Step 1
# -----

# general model
fixed = plm(lcrmrte~lprbarr+lprbconv+lprbpris+lavgsen+lpolpc+ldensity+ltaxpc+lwcon+lwfir,
            data=crime, model="within")
summary(fixed)
```

```
# -----
# Step 2
# -----
# general model without ldensity
fixed = plm(lcrmte~lprbarr+lprbconv+lprbpris+lavgse+lpolpc+ltaxpc+lwcon+lwfir,
            data=crime, model="within")
summary(fixed)

# general model
fixed = plm(lcrmte~lprbarr+lprbconv+lprbpris+lavgse+lpolpc+ldensity+ltaxpc+lwcon+lwfir,
            data=crime, model="within")
summary(fixed)

# ldensity and ltaxpc
h <- rbind(c(0,0,0,0,0,1,0,0,0), c(0,0,0,0,0,0,1,0,0))
wald.test.results = wald.test(b = coef(fixed), Sigma = vcov(fixed), L = h)
wald.test.results

# -----
# Step 3
# -----
# general model without ldensity and ltaxpc
fixed = plm(lcrmte~lprbarr+lprbconv+lprbpris+lavgse+lpolpc+lwcon+lwfir,
            data=crime, model="within")
summary(fixed)

# general model
fixed = plm(lcrmte~lprbarr+lprbconv+lprbpris+lavgse+lpolpc+ldensity+ltaxpc+lwcon+lwfir,
            data=crime, model="within")
summary(fixed)

# ldensity and ltaxpc
h <- rbind(c(0,0,0,0,0,1,0,0,0), c(0,0,0,0,0,0,1,0,0), c(0,0,0,1,0,0,0,0,0))
wald.test.results = wald.test(b = coef(fixed), Sigma = vcov(fixed), L = h)
wald.test.results

# -----
# Step 4
# -----
# general model without ldensity and ltaxpc and lavgse
fixed = plm(lcrmte~lprbarr+lprbconv+lprbpris+lpolpc+lwcon+lwfir,
            data=crime, model="within")
summary(fixed)

# general model
fixed = plm(lcrmte~lprbarr+lprbconv+lprbpris+lavgse+lpolpc+ldensity+ltaxpc+lwcon+lwfir,
            data=crime, model="within")
summary(fixed)
```

```
# ldensity and ltaxpc
h <- rbind(c(0,0,0,0,0,1,0,0,0), c(0,0,0,0,0,0,1,0,0), c(0,0,0,1,0,0,0,0,0),
c(0,0,0,0,0,0,0,0,1))
wald.test.results = wald.test(b = coef(fixed), Sigma = vcov(fixed), L = h)
wald.test.results

# lwfir might be removed from the model

# -----
# Step 5
# -----
# general model without ldensity and ltaxpc and lavgsen and lwfir
fixed = plm(lcrmte~lprbarr+lprbconv+lprbpris+lpolpc+lwcon,
            data=crime, model="within")
summary(fixed)

# general model
fixed = plm(lcrmte~lprbarr+lprbconv+lprbpris+lavgsen+lpolpc+ldensity+ltaxpc+lwcon+lwfir,
            data=crime, model="within")
summary(fixed)

# ldensity and ltaxpc
h <- rbind(c(0,0,0,0,0,1,0,0,0), c(0,0,0,0,0,0,1,0,0), c(0,0,0,1,0,0,0,0,0),
c(0,0,0,0,0,0,0,0,1))
wald.test.results = wald.test(b = coef(fixed), Sigma = vcov(fixed), L = h)
wald.test.results}
```

Contains data from crime_13.dta

obs: 630
vars: 59
size: 128,520

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variable name	storage type	display format	value label	variable label
county	int	%9.0g		county identifier
year	byte	%9.0g		81 to 87
crmrtte	float	%9.0g		crimes committed per person
prbarr	float	%9.0g		'probability' of arrest
prbconv	float	%9.0g		'probability' of conviction
prbpris	float	%9.0g		'probability' of prison sentenc
avgsen	float	%9.0g		avg. sentence, days
polpc	float	%9.0g		police per capita
density	float	%9.0g		people per sq. mile
taxpc	float	%9.0g		tax revenue per capita
wcon	float	%9.0g		weekly wage, construction
wtuc	float	%9.0g		wkly wge, trns, util, commun
wtrd	float	%9.0g		wkly wge, whlesle, retail trade
wfir	float	%9.0g		wkly wge, fin, ins, real est
wser	float	%9.0g		wkly wge, service industry
wmfg	float	%9.0g		wkly wge, manufacturing
wfed	float	%9.0g		wkly wge, fed employees
wsta	float	%9.0g		wkly wge, state employees
wloc	float	%9.0g		wkly wge, local gov emps
mix	float	%9.0g		offense mix: face-to-face/other
pctymle	float	%9.0g		percent young male
lcrmrtte	float	%9.0g		log(crmrtte)
lprbarr	float	%9.0g		log(prbarr)
lprbconv	float	%9.0g		log(prbconv)
lprbpris	float	%9.0g		log(prbpris)
lavgsen	float	%9.0g		log(avgsen)
lpolpc	float	%9.0g		log(polpc)
ldensity	float	%9.0g		log(density)
ltaxpc	float	%9.0g		log(taxpc)
lwcon	float	%9.0g		log(wcon)
lwtuc	float	%9.0g		log(wtuc)
lwtrd	float	%9.0g		log(wtrd)
lwfir	float	%9.0g		log(wfir)
lwser	float	%9.0g		log(wser)
lwmfg	float	%9.0g		log(wmfg)
lwfed	float	%9.0g		log(wfed)
lwsta	float	%9.0g		log(wsta)
lwloc	float	%9.0g		log(wloc)
lmix	float	%9.0g		log(mix)
lpctymle	float	%9.0g		log(pctymle)
lpctmin	float	%9.0g		log(pctmin)

Sorted by: county year